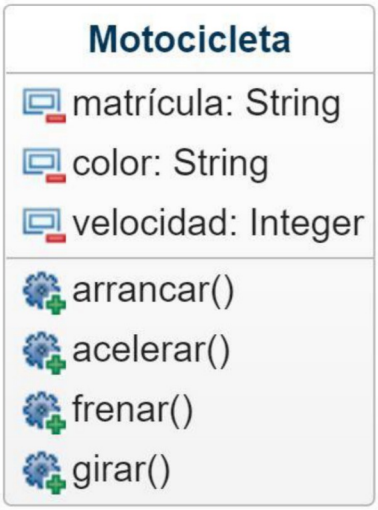
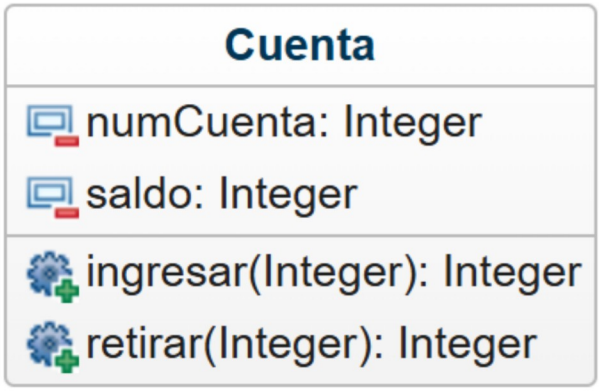
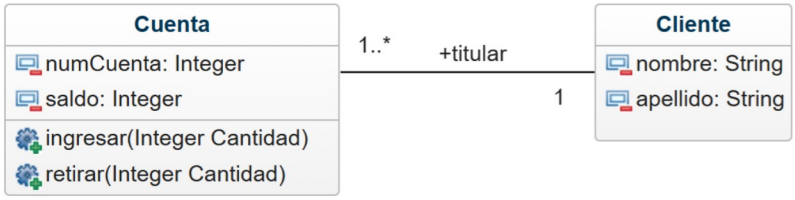
Pas de UML a Java




```
public class Motocicleta {
    private String matricula;
    private String color;
    private Integer velocidad;

    public void arrancar(){}
    public void acelerar(){}
    public void frenar(){}
    public void girar(){}
}
```



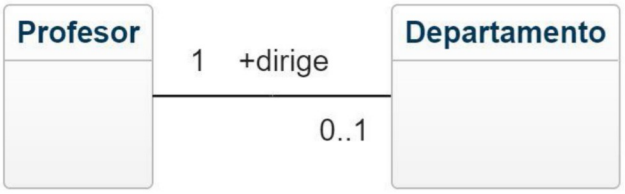

```
public class Cuenta {
    private Integer numCuenta;
    private Integer saldo;

    public Integer ingresar(Integer cantidad){
      saldo = saldo + cantidad;
      return saldo;
    }
    public Integer retirar(Integer cantidad){
      saldo = saldo - cantidad;
      return saldo;
    }
}
```

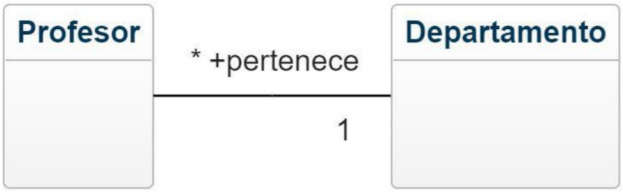


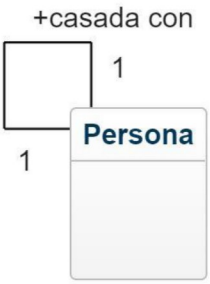

```
public class Cuenta{
    ...
    private Cliente titular;
    ...
}
public class Cliente{
    ...
    private ArrayList<Cuenta> cuentas;
    ...
}
```

Multiplicidad de las asociaciones

Multiplicidad	Significado
1	Uno y sólo uno
0..1	Cero o uno
N..M	Desde N hasta M
*	Cero o varios
0..*	Cero o varios
1..*	Uno o varios (al menos uno)

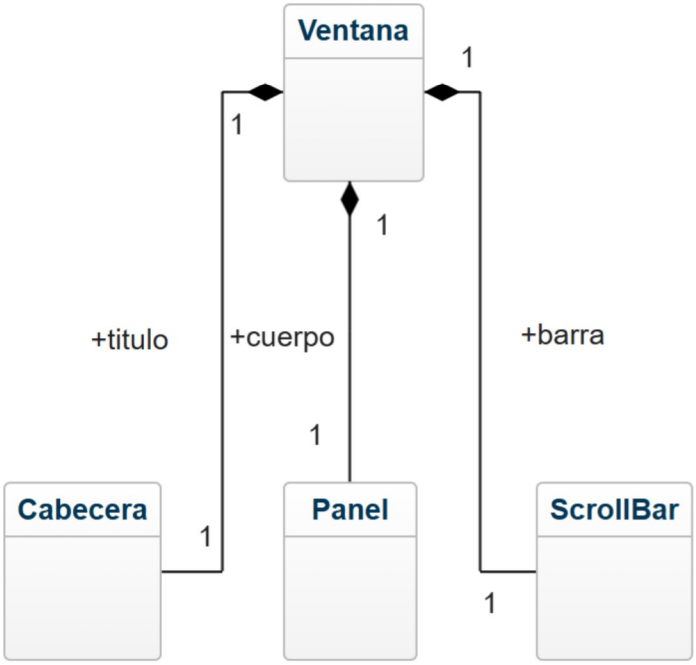






Composición

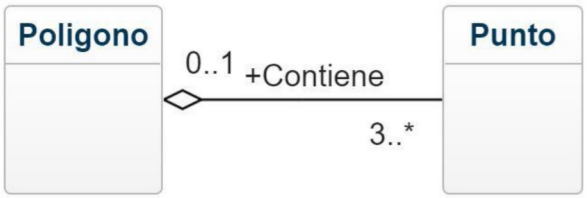
Las partes solo existen asociadas al compuesto




```
Public class Ventana{
    Private Cabecera cabecera;
    Private Panel panel;
    Private ScrollBar scrollBar;

    Public Ventana(){
        this.cabecera = new Cabecera();
        this.panel      = new Panel();
        this.scrollBar  = new ScrollBar();
    }
}
```

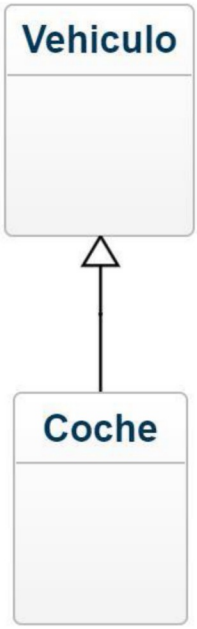
Agregación

Las partes pueden formar parte de distintos agregados.

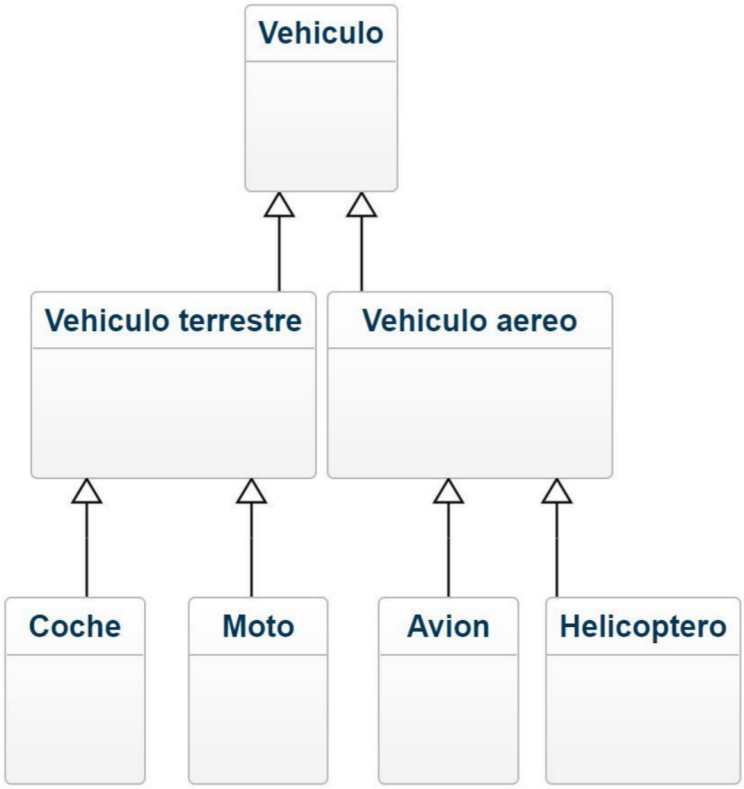



```
Public class Poligono{
    Private ArrayList<Punto> puntos;

    Public Poligono(){}

    Public void dibujar(ArrayList<Punto> puntos){
        this.puntos = puntos;
    }
}
```

Herencia (Generalización / Especialización)

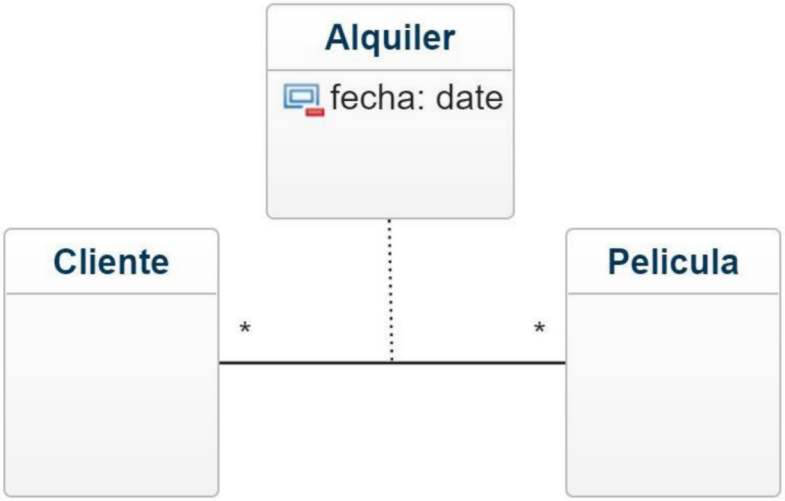



```
Public class Vehiculo(){
}
Public class Coche extends Vehiculo(){
}
```



Clase asociación

La clase asociación se usa para indicar que la asociación existente entre dos clases tiene atributos propios.




```
Public class Cliente(){
...
}
Public class Pelicula(){
...
}
Public class Alquiler(){
    Private Cliente cliente;
    Private Pelicula pelicula;
    Private Date fecha;

    Public Alquiler(Cliente cliente, Pelicula pelicula, Date fecha){
        this.cliente   = cliente;
        this.pelicula = pelicula;
        this.fecha    = fecha;
    }
…
}
```